

NLP Assignment 2 Report

Muhammad Isham

23i-2606

DS-C

<https://github.com/isham78/i23-2606-NLP-Assignment2>

Overview

The work has three parts: (1) embeddings and evaluation, (2) POS and NER tagging, and (3) document classification.

Part 1 Results

Core corpus and vectorization outputs:

- Total documents: 229
- Vocabulary size: 10001
- TF-IDF matrix shape: (229, 10001)
- PPMI shape: (10001, 10001), non-zero values: 716194
- Words in t-SNE: 200
- Analogy accuracy summary: 10/10 correct

MRR values from notebook output:

Condition	MRR
C1 PPMI baseline	0.1491
C2 Skip-gram on raw.txt	0.0312
C3 Skip-gram on cleaned.txt	0.0393
C4 Skip-gram (d=200)	0.0257

Best MRR: C1 PPMI baseline.

Part 2 Results

Data summary:

- Selected anchor topics: international, politics, sports
- Sampled sentence count: 500
- Split sizes: Train 350, Validation 75, Test 75
- Loaded split sizes: 350, 75, 75
- Encoded samples: 350, 75, 75

POS validation results:

- Frozen embeddings best validation macro-F1: 0.6884 (best epoch 30)
- Fine-tuned embeddings best validation macro-F1: 0.9157 (best epoch 22)

POS setting	Test token accuracy	Test macro-F1
Frozen embeddings	0.9068	0.6940
Fine-tuned embeddings	0.9640	0.9179

NER overall test F1:

- NER-CRF overall entity-level test F1: 0.0000
- NER-Softmax overall entity-level test F1: 0.0000

Ablation results:

Ablation	Metric	Value
A1 Unidirectional LSTM only	POS test macro-F1	0.7760
A2 No dropout	POS test macro-F1	0.9513
A3 Random embedding initialization	POS test macro-F1	0.9843
A4 Softmax instead of CRF (NER)	NER entity-level test F1 (overall)	0.0000

Baseline for A1/A2/A3 in notebook: POS macro-F1 = 0.9179. Reference for A4 in notebook: CRF NER overall F1 = 0.0000.

Part 3 Results

Final text classification results:

Model	Test accuracy	Macro-F1
Transformer (ours)	0.7000	0.4465
BiLSTM baseline	0.5000	0.1333

Other comparison numbers:

- Accuracy gap (Transformer - BiLSTM): 0.2000
- Macro-F1 gap (Transformer - BiLSTM): 0.3132
- BiLSTM best validation epoch: 5
- Transformer best validation epoch: 17
- BiLSTM mean epoch time (s): 2.80

Conclusion

Part 1 gives the best retrieval score with the PPMI baseline. Part 2 shows that fine-tuned embeddings improve POS results (test macro-F1 0.9179). NER overall F1 is 0.0000 for both CRF and Softmax in the saved run. Part 3 shows better performance for the Transformer than the BiLSTM baseline on both accuracy and macro-F1.